perceive and understand speech. Words are recognized by their sounds, but also by the *context* of the sentence, and the *expectations* of the listener. For example, imagine hearing the two sentences:

The child wore a spider ring on Halloween.

He was an American spy during the war.

Even if exactly the same sounds were produced to convey the underlined words, listeners *hear* the correct words for the context. From your accumulated knowledge about the world, you know that children don't wear secret agents, and people don't become spooky jewelry during wartime. This usually isn't a conscious act, but an inherent part of human hearing.

Most speech recognition algorithms rely only on the sound of the individual words, and not on their context. They attempt to *recognize words*, but not to *understand speech*. This places them at a tremendous disadvantage compared to human listeners. Three annoyances are common in speech recognition systems: (1) The recognized speech must have distinct pauses between the words. This eliminates the need for the algorithm to deal with phrases that sound alike, but are composed of different words (i.e., *spider ring* and *spy during*). This is slow and awkward for people accustomed to speaking in an overlapping flow. (2) The vocabulary is often limited to only a few hundred words. This means that the algorithm only has to search a limited set to find the best match. As the vocabulary is made larger, the recognition time and error rate both increase. (3) The algorithm must be *trained* on each speaker. This requires each person using the system to speak each word to be recognized, often needing to be repeated five to ten times. This personalized database greatly increases the accuracy of the word recognition, but it is inconvenient and time consuming.

The prize for developing a successful speech recognition technology is enormous. Speech is the quickest and most efficient way for humans to communicate. Speech recognition has the potential of replacing writing, typing, keyboard entry, and the electronic control provided by switches and knobs. It just needs to work a little better to become accepted by the commercial marketplace. Progress in speech recognition will likely come from the areas of artificial intelligence and neural networks as much as through DSP itself. Don't think of this as a technical *difficulty*; think of it as a technical *opportunity*.

Nonlinear Audio Processing

Digital filtering can improve audio signals in many ways. For instance, *Wiener filtering* can be used to separate frequencies that are mainly signal, from frequencies that are mainly noise (see Chapter 17). Likewise, *deconvolution* can compensate for an undesired convolution, such as in the restoration of old